FRESNEL'S BIPRISM

Name: Vedanta Mohapatra

Section-1, Group -1

Roll No: 20CS02001

Date of Experiment: 22/04/2021

Aim of the experiment

To determine the wave length of the given source using Fresnel's biprism.

Apparatus Required

(i) Fresnel's biprism set up

(ii) Optical bench

(iii) LASER source

(iv) White screen with stand

(v) Two lenses with focal length 20 mm & 300 mm

(vi) Measuring tape

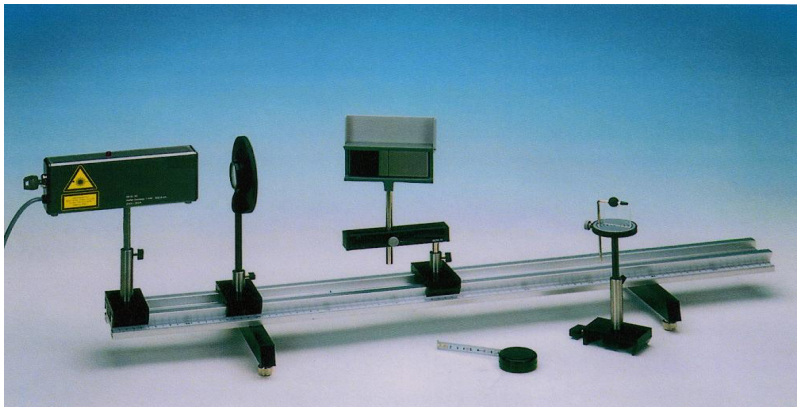


Fig-1: Experimental setup

Theory

A biprism consists of two acute angled prisms placed base to base. The experimental set up and the schematic diagram of a typical biprism is shown in Fig-1 and Fig-2 respectively. A biprism acts as a single prism with an obtuse angle 179^0 and two equal acute angles of $30'$ each. When light falls on the lower part of the biprism it bends upward and appears as if it has come from the virtual source B. Similarly when light fall on the upper part of the biprism, it bends down ward and appears as if it has come from the virtual source A. Thus A and B act as two coherent virtual sources and we get interference fringes of equal width at the screen between E and F.

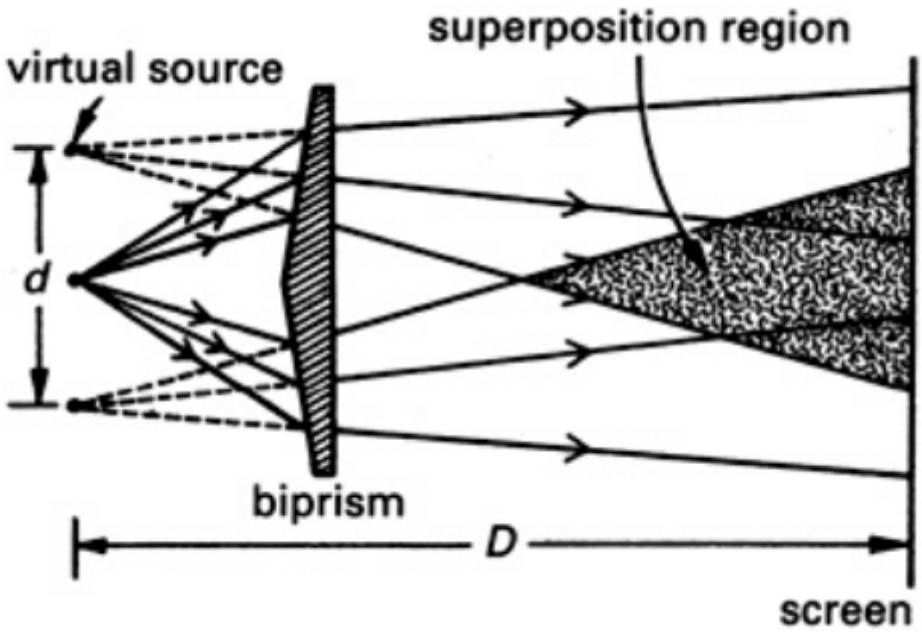


Fig-2: Schematic diagram

From the theory of interference of light we have, $\beta = \frac{\lambda D}{d}$

Where β is the width of the bright or the dark fringe

λ is the wave length of the light used

D is the distance of the screen from the source

d is the distance between the two virtual sources

To measure d, a lens of focal length 300 mm is placed between the biprism and the screen such that the real images of the two virtual sources are formed on the screen and the distance between these real images 'b' is measured.

From basic optics we know that $\frac{b}{d} = \frac{v_1}{u_1}$ or $d = \frac{bu_1}{v_1}$

where u_1 is the distance of the virtual source from the lens and v_1 is the distance of the screen from the lens. The value of u_1 is found out by applying the lens formula

$$u_1 = \frac{Fv_1}{v_1 - F}$$

So we obtain $d = \frac{bF}{(v_1 - F)}$.

Thus obtaining the values of d, β and D the value of λ can be determined by the formula $\lambda = \frac{\beta d}{u_1 + v_1}$.

Procedure

(i) Place the source and the lenses of focal length 20mm at appropriate positions on the optical bench.

(ii) Place the biprism on the prism table and adjust it such that the light from the source falls exactly on the centre of the biprism. Place the screen with a ruler paper fixed on it at an approximate distance of 2 m from the source.

(iii) The screen should be perpendicular to the optical bench. Adjust the prism by rotating it slightly so that a number of parallel fringes of equal width are formed on the screen.

(iv) Measure the width of 10 consecutive bright fringes.

(v) Place the lens of focal length 300 mm on the optical bench. Adjust the position of the lens such that the two real images of the virtual sources are formed on the screen.

(vi) Measure the distance b between the two images of the two virtual sources with the help of a scale.

(vii) Repeat these observations for 4 to 5 times for different positions of the screen.

Observations

Table-1: Position of various optical components

Source Position (cm)	Lens L_1 (cm)	Biprism (cm)	Lens L_2 (cm)	Screen (cm)
3.6	14.5	37.5	54.2	215
3.6	14.5	37.5	57.3	216
3.6	14.5	37.5	58.5	219
3.6	14.5	37.5	59.8	218
3.6	14.5	37.5	60.6	220

Table-2: Measurement of fringe width

Width of 10 fringes (cm)	Fringe width (β) (cm)	Mean fringe width (β) (cm)
1.1	0.11	0.13
1.2	0.12	
1.5	0.15	
1.3	0.13	
1.4	0.14	

Table-3: Measurement of the separation of the virtual sources

Object to lens distance (u_1) (cm)	**Image to lens distance (v_1) (cm)**	**Separation between the real images (b) (cm)**	**Separation between the virtual images** $d = \frac{bF}{(v_1 - F)}$ **(cm)**	**Value of λ (nm)** $= \frac{\beta . d}{u_1 + v_1}$
38.55	135.3	0.5	0.142	1061.8
38.32	138.2	0.45	0.125	920.5
37.91	143.8	0.5	0.132	944.3
37.48	150.4	0.5	0.124	857.9
37.32	152.8	0.4	0.098	670.1

Finally, λ = Average of λ calculated = $\frac{1061.8+920.5+944.3+857.9+670.1}{5}$ = 890.9 nm

Conclusions

(i) Interference pattern due to two virtual sources of light is observed. Wave nature of light is confirmed. From the experimental data obtained, the wavelength of light was calculated. It serves as a nice proof of concept.

(ii) Some limitations faced by this experiment are need of monochromatic light of suitable wavelength: for instance, the experiment would not work for normal light. Also, adjusting the prism for obtaining a clean interference patter is very hard. Width of fringes observed is actually not constant and distortion of fringes becomes more and more as we travel farther from 0th order maxima.

(iii) Although the experiment (as described here) does not find many real world applications, the concepts which come into play here have tremendous applications. Interference and diffraction are two sides of

the same coin. The both come into play when there are various close sources and obstructions (having dimensions of the order of the wavelength of light used). As such, they are kept in mind when designing telescopes, interferometers, diffraction gratings, making precision measurements(such as detection of gravitational waves by LIGO) and has various other applications in modern science.

Precautions

(i) One should not look directly into LASER light because it may damage the eye.
(ii) The optical bench should be levelled properly.
(iii) The refracting edge should be normal to the direction of the light.
(iv) The width of the fringe should be accurately measured.
(v) As many fringes as possible should be taken to find out the mean fringe width β.

Considerations

(i) What is interference of light and what are the different ways in which light interferes?

Interference can be defined as the superposition of light coming from various close (separation of the order of wavelength) monochromatic coherent sources. Due to the wave character of light, the light may interfere constructively (constructive interference) and the amplitude/brightness of a region is greater than its surrounding regions. Or the light may interfere destructively and cause the brightness of certain regions to be low.

(ii) What is wave theory of light?

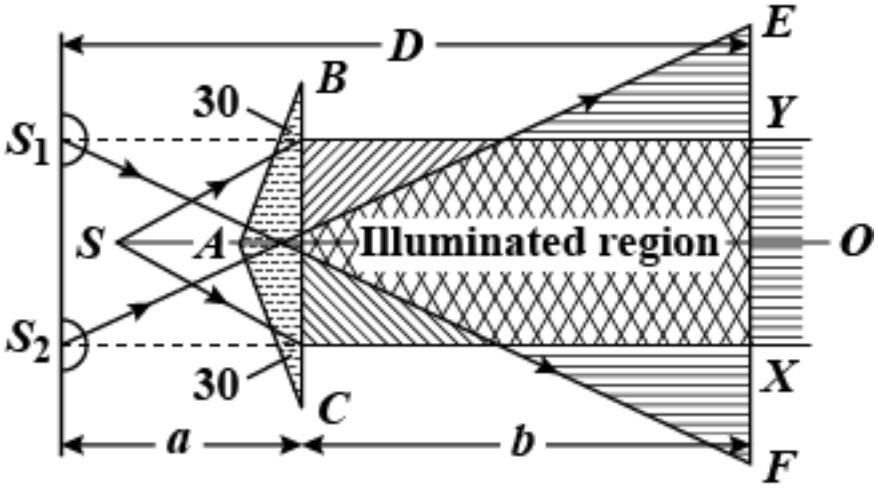
Wave theory of light treats light as an electromagnetic wave and various properties like superposition theorem, huygen's principle holds for waves and therefore are valid for light waves, which explains various phenomenon like snell's law ,YDSE, diffraction, etc. which could not be explained previously be newton's particle model of light.

(iii) Is the Fresnel's biprism experiment better than Young's double slit experiment? Justify your answer.

For the Young's slits experiment, we must approximate that the slits act as point sources. This however is not the case, since the slits have finite width. This finite size of the secondary slits gives rise to unwanted diffraction effects which causes errors. The Fresnel biprism overcomes this problem of extended secondary slits by replacing them with virtual slits which are point-like

(iv) What are coherent sources and how coherent sources are realized in biprism experiment?

Two light sources are said to be coherent if they are emitting light of same wavelength (or same frequency) such that the originating phase difference between them is either zero or has a fixed constant value. In the biprism experiment, the light emerging from the slit fall on the biprism. The edge A of the biprism divides the incident wave front into two parts. One is through upper half AB of biprism and appears to coming from virtual source S_1. Other is from lower half AC of biprism and appears to coming virtual S_2 The interference fringes are seen in the overlapping region XY and can be seen by eyepiece.



(v) What will happen if we increase the focal length of the lens?

The distance between slit and eyepiece must be greater than four times the focal length of the convex lens (D>4f) while measuring d. So, increasing the focal length without altering the distance between the slit and the screen might result in D<4f, which would hinder the experiment as real images wouldn't be obtained on the screen. Otherwise, if the this condition is not violated, we would see no change in the values of d and λ.